

Spherical Harmonics and Polarizations

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This note goes through the connections between scalar, vector, tensor spherical harmonics, spin-weighted spherical harmonics, and the polarization of gravitational waves and the CMB.

Spherical Harmonics

Scalar Spherical Harmonics

First we define the normal scalar spherical harmonics.

$$Y_{\ell m}(\theta, \phi) = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}} P_{\ell m}(\cos \theta) e^{im\phi} \quad (1)$$

where $P_{\ell m}(\cos \theta)$ are the Legendre polynomials. This is the convention of *Mathematica*. In this note, I won't list any of the orthogonality properties, but they exist for these and the subsequent harmonics. They just don't really matter a whole lot for this understanding. The scalar spherical harmonics also have the subsequent properties

$$\overline{Y_{\ell m}}(\theta, \phi) = (-1)^m Y_{\ell(-m)}(\theta, \phi) \quad (2)$$

$$Y_{\ell m}(\pi - \theta, \phi + \pi) = (-1)^\ell Y_{\ell m}(\theta, \phi). \quad (3)$$

Vector Spherical Harmonics

Next we define the vector spherical harmonics. We take these definitions from Baumgarte and Shapiro [1]. Vector spherical harmonics form a basis on the 2-sphere, S^2 . Therefore, there needs to be two linearly independent vectors to form this basis. The way we construct this basis is by taking an gradient-like derivative and a curl-like derivative of the scalar spherical harmonics.

$$E_{\mu}^{\ell m}(\theta, \phi) = \nabla_{\mu} Y_{\ell m}(\theta, \phi) \quad (4)$$

$$S_{\mu}^{\ell m}(\theta, \phi) = \epsilon_{\mu}^{\nu} \nabla_{\nu} Y_{\ell m}(\theta, \phi) \quad (5)$$

In this equation, ϵ_{μ}^{ν} is the 2D Levi-Cevita tensor on S^2 , given by

$$\epsilon_{\mu\nu} = \begin{pmatrix} 0 & \sin \theta \\ -\sin \theta & 0 \end{pmatrix}. \quad (6)$$

Here, we should also note that the metric on S^2 is given by

$$\sigma_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}, \quad \sigma^{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \csc^2 \theta \end{pmatrix}. \quad (7)$$

With these definitions, you can also show that $E_{\mu}^{\ell m}$ and $S_{\mu}^{\ell m}$ are orthogonal,

$$E_{\ell m}^{\mu} S_{\mu}^{\ell m} = \sigma^{\mu\nu} E_{\nu}^{\ell m} S_{\mu}^{\ell m} = 0. \quad (8)$$

Just like the scalar spherical harmonics, the vector spherical harmonics have similar properties, namely

$$\overline{E_\mu^{\ell m}}(\theta, \phi) = (-1)^m E_\mu^{\ell(-m)}(\theta, \phi) \quad (9)$$

$$\overline{S_\mu^{\ell m}}(\theta, \phi) = (-1)^m S_\mu^{\ell(-m)}(\theta, \phi) \quad (10)$$

$$E_{\mu'}^{\ell m}(\theta' = \pi - \theta, \phi' = \phi + \pi) = (-1)^\ell E_\mu^{\ell m}(\theta, \phi), \quad (11)$$

$$S_{\mu'}^{\ell m}(\theta' = \pi - \theta, \phi' = \phi + \pi) = -(-1)^\ell S_\mu^{\ell m}(\theta, \phi). \quad (12)$$

These last two properties are particularly important because they show how this basis of vector spherical harmonics corresponds to an even/polar/electric and odd/axial/magnetic parity. All these terms are basically interchangeable, but we'll stick to even and odd unless clear reason to otherwise. The $E_\mu^{\ell m}(\theta, \phi)$ have even parity under spatial inversion, while $S_\mu^{\ell m}(\theta, \phi)$ have odd parity under spatial inversion. What this means is that any vector field on S^2 can be decomposed into its even and odd parity components which are linearly independent of one another. One thing to note in these transformations, you're not only taking $(\theta' = \pi - \theta, \phi' = \phi + \pi)$, but you're also changing the basis vectors. We have that $(\hat{e}_{\theta'} = -\hat{e}_\theta, \hat{e}_{\phi'} = \hat{e}_\phi)$. Thus when you do the transformation, you need to take $(\theta \rightarrow \theta', \phi \rightarrow \phi')$ as well as multiply by the proper factors of the Jacobian. It's this transformation of the basis vectors that really give these things the proper spin weights. Scalar values have spin-weight 0 because they have no basis vectors. Vector values has spin-weight ± 1 because they have one basis vector. Tensor values have spin-weight ± 2 because they have two basis vectors. This means if you rotate the basis vector by some angle ψ via $e^{i\psi}$, each of these objects will transform as $e^{is\psi}$.

Tensor Spherical Harmonics

Tensor spherical harmonics carry on the same spirit as the vector harmonics, just adding another derivative in their definitions. Note that the basis we define here is only a complete basis for symmetric, traceless, rank-2 tensors. This exactly the form the metric perturbation take in the transverse-traceless gauge, and also the structure of the polarization tensor for CMB stuff. In terms of the vector and scalar spherical harmonics, the tensor spherical harmonics are defined as

$$Z_{\mu\nu}^{\ell m}(\theta, \phi) = \nabla_\mu E_\nu^{\ell m} + \frac{\ell(\ell+1)}{2} \sigma_{\mu\nu} Y^{\ell m} = \nabla_\mu \nabla_\nu Y^{\ell m} + \frac{\ell(\ell+1)}{2} \sigma_{\mu\nu} Y^{\ell m} \quad (13)$$

$$X_{\mu\nu}^{\ell m}(\theta, \phi) = \nabla_{(\mu} S_{\nu)}^{\ell m} = \nabla_{(\mu} \epsilon_{\nu)}^\rho \nabla_\rho Y^{\ell m}. \quad (14)$$

Just like the vector spherical harmonics, these are also orthogonal, and they are also traceless.

$$X_{\ell m}^{\mu\nu} Z_{\mu\nu}^{\ell m} = 0 \quad (15)$$

$$\sigma^{\mu\nu} S_{\mu\nu}^{\ell m} = 0, \quad \sigma^{\mu\nu} Z_{\mu\nu}^{\ell m} = 0. \quad (16)$$

Once again, these have the set of properties

$$\overline{Z_\mu^{\ell m} \nu}(\theta, \phi) = (-1)^m Z_\mu^{\ell(-m)} \nu(\theta, \phi) \quad (17)$$

$$\overline{X_\mu^{\ell m} \nu}(\theta, \phi) = (-1)^m X_\mu^{\ell(-m)} \nu(\theta, \phi) \quad (18)$$

$$Z_{\mu'\nu'}^{\ell m}(\theta' = \pi - \theta, \phi' = \phi + \pi) = (-1)^\ell Z_{\mu\nu}^{\ell m}(\theta, \phi), \quad (19)$$

$$X_{\mu'\nu'}^{\ell m}(\theta' = \pi - \theta, \phi' = \phi + \pi) = -(-1)^\ell X_{\mu\nu}^{\ell m}(\theta, \phi). \quad (20)$$

Spin-Weighted Spherical Harmonics

In this section, we draw the connection between the scalar, vector, tensor spherical harmonics with the spin-weighted spherical harmonics. Roughly speaking, spin-weighted spherical harmonics

correspond to a projection of the scalar/vector/tensor spherical harmonics onto a tetrad basis of S^2 . For the tetrad basis, we need two tetrad legs m_μ and $\bar{m}_\mu$ that satisfy the conditions

$$m_\mu \bar{m}^\mu = 1 \quad (21)$$

$$\sigma_{\mu\nu} = m_{(\mu} \bar{m}_{\nu)}. \quad (22)$$

The most natural choice for this on S^2 is

$$m_\mu = \frac{1}{\sqrt{2}}(1, i \sin \theta), \quad \bar{m}_\mu = \frac{1}{\sqrt{2}}(1, -i \sin \theta). \quad (23)$$

With the tetrad defined, we can project and define our spin-weighted spherical harmonics, ${}_s Y_{\ell m}(\theta, \phi)$.

For spin-weight 0, the scalar spherical harmonics have nothing to project, so we have

$${}_0 Y_{\ell m}(\theta, \phi) = Y_{\ell m}(\theta, \phi). \quad (24)$$

For spin-weight ± 1 , we project out the singular index, yielding

$${}_{+1} Y^{\ell m}(\theta, \phi) \propto m^\mu (E_\mu^{\ell m} - i S_\mu^{\ell m}) \quad (25)$$

$${}_{-1} Y^{\ell m}(\theta, \phi) \propto \bar{m}^\mu (E_\mu^{\ell m} + i S_\mu^{\ell m}). \quad (26)$$

We should pause here and think about why the spin-weighted harmonics have this particular structure. We want the spin weighted spherical harmonics to have definite spin weights. However, the E and S vector spherical harmonics mix both $s = +1$ and $s = -1$ spin weights. Thus, we need to separate these out. To get a definite spin weight, we need something circular in a sense. Normally the way we do this is with something like $\hat{x} \pm i\hat{y}$ or $\cos\theta \pm i \sin\theta$. You can almost think of E and S being parity eigenstates while ${}_{\pm 1} Y^{\ell m}$ are helicity eigenstates. This also means that the spin-weighted spherical harmonics correspond to circular polarization, with $s = -1$ the right-handed polarization, and $s = +1$ the left handed. Note that there are annoying proportionality constants out in front of this to ensure proper normalization, but these aren't necessary to understand the connection between all these concepts. The above concept that E and S mix spin weights can be further illustrated by inverting the relationships, giving

$$E_\mu^{\ell m} \propto m_\mu {}_{-1} Y^{\ell m} + \bar{m}_\mu {}_{+1} Y^{\ell m} \quad (27)$$

$$S_\mu^{\ell m} \propto m_\mu {}_{-1} Y^{\ell m} - \bar{m}_\mu {}_{+1} Y^{\ell m} \quad (28)$$

The general concepts continue for $s = \pm 2$.

$${}_{+2} Y^{\ell m}(\theta, \phi) \propto m^\mu m^\nu (Z_{\mu\nu}^{\ell m} - i X_{\mu\nu}^{\ell m}) \quad (29)$$

$${}_{-2} Y^{\ell m}(\theta, \phi) \propto \bar{m}^\mu \bar{m}^\nu (Z_{\mu\nu}^{\ell m} + i X_{\mu\nu}^{\ell m}). \quad (30)$$

The inverted relationships are

$$Z_{\mu\nu}^{\ell m} \propto m_\mu m_\nu {}_{-2} Y^{\ell m} + \bar{m}_\mu \bar{m}_\nu {}_{+2} Y^{\ell m} \quad (31)$$

$$X_{\mu\nu}^{\ell m} \propto m_\mu m_\nu {}_{-2} Y^{\ell m} - \bar{m}_\mu \bar{m}_\nu {}_{+2} Y^{\ell m}. \quad (32)$$

It's good to once again point out here that ${}_{+2} Y^{\ell m}$ corresponds to right-handed circular polarization and ${}_{-2} Y^{\ell m}$ to left-handed circular polarization.

Contextualized to CMB and GWs

So we've done all this work to understand how all these concepts connect to one another, but it's still important to connect them back to the real world. This whole exploration was started by looking at the polarization of imprint primordial GWs leave on the CMB afterall. We'll talk about that in

a minute, but first let's talk about GWs. So for gravitational wave detections, we normally consider plus and cross polarizations, h_+ and $h_\times$. In terms of NR extraction, they're given the $h_{\ell m}$ of

$$h_+ - ih_\times = \sum_{\ell m} h_{\ell m} {}_{-2}Y^{\ell m}. \quad (33)$$

They then can just take the real and imaginary parts to get what's seen in the detector frame. Note that plus and cross polarization only naturally fall out when we're in transverse-traceless gauge, allowing use to use the tensor spherical harmonics we derived previously. This convention is basically just picking off the left handed polarization of the GW. Likewise, we could pick off the right handed polarization,

$$h_+ + ih_\times = \sum_{\ell m} \tilde{h}_{\ell m} {}_{+2}Y^{\ell -m}. \quad (34)$$

Here, $\tilde{h}_{\ell m}$ is the conjugate of $h_{\ell m}$ up to a minus sign. The one thing I'm not sure about here, is I feel like I should be able to express this in terms of $+m$ rather than $-m$, but I'm not sure. Either way, what we're doing here makes sense. I express circular polarization in terms of the spin-weighted $s = \pm 2$ harmonics, and then within that, we can pick off the plus and cross polarization. You may be inclined to think that the plus and cross polarization are exactly the projection of the even and odd parity tensor harmonics, but that's not quite true. If you just go through the math, you'll see you can't express either h_+ or $h_\times$ in terms of just Z or X . I think the way to think about it is just they are the real and imaginary parts of the spin-weighted harmonics.

For the CMB, rather than plus and cross polarization, they think of things in terms of E and B modes. These E and B modes are exactly the Z and X tensor spherical harmonics we've talked about. I'm not sure how they actually detect these polarizations so maybe they do something like GW plus and cross polarization, but I'm not sure. They reason they do this decomposition is because the CMB is predominantly E modes. The things that cause polarization have even parity, meaning they don't have a handedness. This is mostly Thompson/Compton scattering. However, primordial GWs do have a handedness and can produce B modes as we see from the fact that they have both plus and cross polarizations. So if we detect B modes, it might be primordial GWs. There's some more details about the polarization tensor they use, but I won't get into that here.

References

- [1] Thomas W Baumgarte and Stuart L Shapiro. Numerical Relativity: Solving Einstein's Equations on the Computer.